

SI231B: Matrix Computations, 2025 Fall

Homework Set #4

Acknowledgements:

- 1) Deadline: **2025-12-13 23:59:59**
 - 2) Please submit the PDF file to [gradescope](#). Course entry code: N2382J.
 - 3) You have 5 “free days” in total for all late homework submissions.
 - 4) If your homework is handwritten, please make it clear and legible.
 - 5) All your answers are required to be in English.
 - 6) Write down the major steps for deriving the solution; otherwise you may loss points.
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Problem 1. (Quadratic Form) (20 points)

Consider the following function:

$$f(\mathbf{x}) = 2x_1^2 + 2x_2^2 + ax_3^2 + 2x_1x_2 + 2x_1x_3 + 2x_2x_3$$

- 1) Find a symmetric matrix $\mathbf{A} \in \mathbb{R}^{3 \times 3}$ such that $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$. (5 points)
- 2) Find the condition of a such that $\mathbf{A}$ is PSD. (5 points)
- 3) If $a = 2$, find the PSD square root of $\mathbf{A}$. (10 points)

(Hint: You might need to use Gram-Schmidt to find orthonormal eigenvectors.)

Solution:

Problem 2. (Positive Semidefinite Matrices) (20 points)

Let $\mathbf{A} = \mathbf{I} + \boldsymbol{\alpha}\boldsymbol{\alpha}^T$ be an $n \times n$ real symmetric matrix, where $\boldsymbol{\alpha} = [a_1, a_2, \dots, a_n]^T$ is a non-zero real vector, and denote $\beta = \boldsymbol{\alpha}^T \boldsymbol{\alpha} > 0$.

- 1) Prove that $\mathbf{A}$ is a symmetric positive definite matrix. (5 points)
- 2) Find all eigenvalues of $\mathbf{A}$ and the corresponding linearly independent eigenvectors (specify the algebraic multiplicity of each eigenvalue). (5 points)
- 3) Let $\mathbf{C} = (\mathbf{A} - \mathbf{I})^3$. Find the positive definite square root $\mathbf{C}^{1/2}$ of $\mathbf{C}$. (10 points)

Solution:

Problem 3. (Application of Schur Complement)

(10 points)

Suppose $\mathbf{X}$ is the 6×6 matrix:

$$\mathbf{X} = \begin{bmatrix} 2 & 2 & 1 & 0 & 0 & -1 \\ 2 & a & 2 & 0 & 0 & 2 \\ 1 & 2 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 \\ -1 & 2 & 0 & 0 & 0 & 2 \end{bmatrix}$$

Find the condition of a such that $\mathbf{X}$ is PSD. (10 points)

Solution:

Problem 4. (Matrix Inequality, Matrix Norm) (25 points)

- 1) Let $\mathbf{X} \in \mathbb{R}^{n \times r}$ with $r \leq n$, and suppose that

$$\|\mathbf{X}^\top \mathbf{X} - \mathbf{I}_r\|_2 = \tau < 1.$$

Show that the smallest singular value of $\mathbf{X}$ satisfies

$$\sigma_{\min}(\mathbf{X}) \geq 1 - \tau.$$

(Hint: $\|\mathbf{A}\|_2 = \max_{\|\mathbf{z}\|_2=1} \|\mathbf{Az}\|_2 = \max_{\|\mathbf{z}\|_2=\|\mathbf{y}\|_2=1} |\mathbf{y}^\top \mathbf{Az}|$) (15 points)

- 2) Assume matrix $E = ab^\top$, where $a \in \mathbb{R}^m, b \in \mathbb{R}^n, a \neq \mathbf{0}_m, b \neq \mathbf{0}_n$, Prove the maximum singular value $\sigma_1(E) = \|a\|_2 \|b\|_2$. (10 points)

Solution:

Problem 5. (SVD) (15 points)

Let θ_u and θ_v be real numbers that satisfy $0 \leq \theta_v - \theta_u \leq \frac{\pi}{2}$ and

$$A = \begin{bmatrix} \cos(\theta_u) & \sin(\theta_u) \\ \cos(\theta_v) & \sin(\theta_v) \end{bmatrix}.$$

Find the singular value decomposition of A . (Hint: $\cos^2\left(\frac{\theta}{2}\right) = \frac{1+\cos(\theta)}{2}$, $\sin^2\left(\frac{\theta}{2}\right) = \frac{1-\cos(\theta)}{2}$. The sum-to-product and product-to-sum identities are as follows.

$$\sin A + \sin B = 2 \sin\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right),$$

$$\sin A - \sin B = 2 \cos\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right),$$

$$\cos A + \cos B = 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right),$$

$$\cos A - \cos B = -2 \sin\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right).$$

$$\sin \alpha \cos \beta = \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)],$$

$$\cos \alpha \sin \beta = \frac{1}{2} [\sin(\alpha + \beta) - \sin(\alpha - \beta)],$$

$$\cos \alpha \cos \beta = \frac{1}{2} [\cos(\alpha + \beta) + \cos(\alpha - \beta)],$$

$$\sin \alpha \sin \beta = -\frac{1}{2} [\cos(\alpha + \beta) - \cos(\alpha - \beta)].$$

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Solution:

Problem 6. (Sensitivity in Linear Systems)

(10 points)

Consider the least squares problem with a full column rank matrix $A \in \mathbb{R}^{m \times n}$ ($m \geq n$):

$$\min_{x \in \mathbb{R}^n} \|Ax - y\|_2^2.$$

Let Δy be a perturbation (disturbance) added to the observation vector y , and denote the resulting perturbation in the solution by Δx . Since A has full column rank, the least-squares solution is unique and given by $x = A^\dagger y$, where $A^\dagger = (A^\top A)^{-1} A^\top$ is the MoorePenrose pseudoinverse. Consequently,

$$\Delta x = A^\dagger \Delta y.$$

- 1) Derive an inequality that relates the relative error in the solution to the relative error in the observation, i.e., prove

$$\frac{\|\Delta x\|_2}{\|x\|_2} \leq \kappa_2(A) \frac{\|\Delta y\|_2}{\|y\|_2}.$$

Where $\kappa_2(A) = \|A\|_2 \|A^\dagger\|_2$. (Hint: Under the standard assumption in sensitivity analysis that the residual is small, we have $\|Ax\|_2 \approx \|y\|_2$, and thus

$$\frac{1}{\|x\|_2} \lesssim \frac{\|A\|_2}{\|y\|_2}.$$

) (7 points)

- 2) Now let $A = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix}$, where $A_1 \in \mathbb{R}^{n \times n}$ is nonsingular and $A_2 \in \mathbb{R}^{(m-n) \times n}$. Show that the upper bound of $\|\Delta x\|_2$ cannot infinitely increase with the increase of m . (Hint: $\|\Delta x\|_2 = \|A^\dagger \Delta y\|_2$, and use the inequality $\|A^\dagger\|_2 \leq \|A_1^{-1}\|_2$ to explain.) (3 points)

Solution: